

Embolisation treatments for liver cancer

Embolisation is a way of blocking the blood flow to the cancer in the liver. It is often given in combination with chemotherapy or radiation.

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What is embolisation?

Embolisation is a way of blocking the blood flow to the cancer in [the liver](#). A substance is injected into a blood vessel in the liver that goes to the cancer. This reduces the supply of oxygen and energy to the tumour, which can make it shrink or stop growing.

This treatment is used for [primary liver cancer](#) (also called hepatocellular carcinoma or HCC). It can also be used to treat a [secondary cancer in the liver](#).

Embolisation on its own is called trans-arterial embolisation (TAE). But it is often given in combination with either:

- [chemotherapy](#) given into a blood vessel going to the liver – this is called chemoembolisation or trans-arterial chemoembolisation (TACE)
- a type of radiation called radioembolisation or [selective internal radiotherapy \(SIRT\)](#).

Sometimes a type of embolisation called portal vein embolisation (PVE) might be done before a liver resection.

Getting support

We understand that having treatment can be a difficult time for people. We are here to support you. If you want to talk, you can:

- Call the Macmillan Support Line for free on [0808 808 00 00](tel:0808 808 00 00).
- [Chat to our specialists online](#).
- Visit our [liver cancer forum](#) to connect with people who have been affected by liver cancer, share your experience, and ask your questions.

When are chemoembolisation and trans-arterial embolisation used?

For primary liver cancer (HCC)

These treatments may be used when:

- the cancer is only in the liver
- the cancer cannot be removed with surgery or treated with ablation.

Sometimes they might be used before surgery or ablation.

For secondary liver cancer

These treatments are not commonly used to treat secondary liver cancer. Some people might have:

- Chemoembolisation (TACE) to treat secondary liver cancer that has spread from the bowel or a [neuroendocrine tumour \(NET\)](#) in the liver.
- Trans-arterial embolisation (TAE) to shrink a cancer that has spread to the liver from a NET tumour.

Embolisation treatments are not suitable for everyone. They may be given as part of a [clinical trial](#).

Chemoembolisation (TACE) for liver cancer

A chemotherapy drug is injected into a blood vessel going to the liver. This means the tumour gets a stronger dose of the drugs.

You then have an injection of an embolising substance into the blood vessels (arteries) that carry blood to the liver. This blocks the arteries and cuts off the blood supply to the tumour (embolisation).

The chemotherapy drug most often used is [doxorubicin](#). Some people might have another treatment of TACE several weeks later.

How chemoembolisation is given

Preparing for chemoembolisation

You may need to stay in hospital for 1 to 2 nights. Before the treatment, the nurse or doctor usually gives you a mild sedative to help you relax. They then inject some local anaesthetic into the skin at the top of your leg (your groin) to numb the area.

Having treatment

After this, the doctor makes a tiny cut in the skin. They put a fine tube called a catheter through the cut and into a blood vessel called an artery in your groin. The doctor then passes the catheter along the artery until it reaches the blood vessel that takes blood to the liver and tumour. This is called the hepatic artery.

They also put a dye into the blood vessel through the catheter. This shows the blood supply on an [x-ray](#) and shows the doctor where the catheter is. This is called an angiogram.

After this, the doctor injects the chemotherapy drugs or tiny beads coated with chemotherapy directly into the blood vessel going to the tumour. You usually then have an injection of an embolising substance to block this blood vessel. This cuts off the blood supply to the tumour or tumours in the liver (embolisation).

DEB-TACE

Some hospitals use beads that gradually release chemotherapy and also block the blood vessels. This is called DEB-TACE. DEB is short for drug-eluting bead.

The chemotherapy drug [irinotecan](#) may be given this way to treat bowel cancer that has spread to the liver. This treatment is called DEBIRI-TACE.

How trans-arterial embolisation (TAE) is given

The procedure for TAE is the same as for chemoembolisation except you do not have chemotherapy into the liver.

Side effects of chemoembolisation

Chemoembolisation can cause side effects such as:

- a high temperature and flu-like symptoms
- [pain](#) in the upper right side of the tummy area (abdomen)
- [feeling sick \(nausea\)](#)
- feeling very [tired \(fatigue\)](#).

You will be given anti-sickness drugs and painkillers until the side effects get better. This usually takes 1 to 2 weeks.

It is unusual for chemotherapy given in this way to cause side effects outside of your liver. Serious complications are rare, but sometimes it can damage the liver.

Radioembolisation (SIRT) for liver cancer

Radioembolisation uses radioactive beads to destroy cancer cells as well as the small blood vessels in and around the cancer. Destroying the blood supply may help shrink the cancer.

Radioembolisation is sometimes called selective internal radiotherapy (SIRT) or trans-arterial radioembolisation (TARE).

SIRT for primary liver cancer

For [primary liver cancer](#) this treatment can be used if the liver cancer cannot be removed by surgery and:

- your liver is working well ([Child-Pugh grade A](#))
- other treatments such as chemoembolisation (TACE) or targeted and immunotherapy drugs are not suitable.

If there is only 1 large tumour in the liver, SIRT might shrink it so that surgery to remove it is possible. It may take months, a year or longer before SIRT has had its maximum effect.

SIRT might be used instead of TACE if there is only 1 large tumour in the liver. If a tumour is large and has spread into blood vessels, SIRT might be used instead of targeted or immunotherapy drugs.

SIRT for secondary liver cancer

It may be used to treat secondary liver cancer that has spread from the bowel, if the secondary cancer:

- cannot be removed with [surgery](#)
- does not respond to [chemotherapy](#).

You may have SIRT if chemotherapy and its side effects are causing you problems.

How SIRT is given

Preparing for SIRT

You have different tests before the treatment. This includes an [angiogram](#), which is an x-ray that checks the blood vessels. You usually have this up to 2 weeks before SIRT. It shows the doctor where the SIRT beads should go when they are injected.

You usually stay in hospital overnight to have SIRT. You are awake during SIRT. You might have medication to help you to relax and feel sleepy before it.

Having treatment

You have SIRT through a fine tube (catheter). Your doctor puts the catheter into an artery in the top of your leg (groin). They guide it through the artery into a blood vessel that takes blood to the liver. They then inject tiny radioactive beads (microspheres) through the catheter into this blood vessel.

The beads stick permanently in the small blood vessels in and around the liver tumour. They also give off radiation. This damages the cancer cells and destroys the small blood vessels in and around the tumour. Without a blood supply, tumours shrink and may die.

The radiation from each bead only affects tissue nearby. This reduces the risk of damage to healthy cells. The beads lose their radiation quickly. They stay in the liver permanently but are harmless.

After treatment

There are some precautions you will need to take for a short while after SIRT. These are to protect other people around you. Your doctors and nurses will explain this to you.

Side effects of SIRT

Side effects can last for a few days and include:

- a high temperature and flu-like symptoms
- tummy pain
- feeling sick (nausea).

Your doctor will prescribe drugs to control these side effects until they go away. Serious complications are rare.

It takes about 1 to 4 months for the tumour to shrink.

We have more information about [SIRT](#).

Portal vein embolisation (PVE)

The main blood vessel that carries blood to the liver is called the portal vein. A portal vein embolisation (PVE) is a procedure to block a branch of this vein.

This blocks the blood flow to the part of liver to be removed. The blood flow is redirected to the healthy part of the liver and encourages it to grow. This can make sure there is enough liver left behind after surgery for it to work properly.

You usually have a PVE in the x-ray department. You have a local anaesthetic injection to numb an area of your tummy (abdomen). The doctor uses an [ultrasound scan](#) to find the area of the portal vein that supplies blood to the part of the liver with the tumour(s).

They make a small cut in the skin just below the ribcage on the right side of the tummy. Then they gently push a fine tube (catheter) into the portal vein. When the catheter is in place, the doctor injects special glue or very small metal coils into it. This blocks the blood supply to that part of the liver.

PVE usually takes 1½ to 2 hours. You may feel some gentle pushing as the doctor inserts the catheter. Tell them if you feel any pain or discomfort, so they can give you painkillers. You usually stay in hospital overnight after this treatment.

Some people might have a PVE before a [liver resection](#). It is usually done when surgery is possible but there might not be enough remaining liver left afterwards to work properly.

If the liver has grown enough after a PVE, you have a liver resection operation about 3 to 6 weeks later. You may have more scans of your liver during this time.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our primary liver cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Get in touch



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